Comparison of Erasure-Aware Error Correction Decoding for Fault Tolerant Quantum Computing

University of California, Berkeley
C191A Introduction to Quantum Computing I
Final Project. Fall 2025

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Last revised:
November 28, 2025

Abstract

TODO

The abstract should summarize our main points: purpose, methods, results, and conclusions. Keep it short.

Contents

1 Introduction

Motivation & research question

Motivation: Fault-tolerate quantum computation at scale remains a formidable challenge, but we're excited about these recent major advancements:

Decoding with erasure information dramatically increases the surface code error threshold from 0.937% to 4.15%, with an estimated 98% of errors convertible into erasures. [1]

Machine-learned error decoding for quantum processors (surface code) by Bausch et al uses a transformer-based neutral network to decode error data with realistic noise models, including qubit cross-talk and state leakage.

"Requires a system capable of dissipatively removing physical errors while simultaneously performing arbitrary coherent manipulation of the encoded logical qubits." [2]

Sub-questions & hypotheses

2 Background & Related Work

2.1 Noise Models

Pauli Noise. The simplest model used to evaluate Quantum Error Correction (QEC) assumes all physical qubit errors can be represented by uniformly random Pauli operators applied during circuit simulation. In our simulations we implement a depolarizing error channel to represent Pauli noise, which has the following effect on some quantum state ρ with p probability of becoming a maximally-mixed state and $(1-p)$ probability of remaining unechanged:

$$E_{\text{pauli}}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \qquad \text{[Depolarizing error channel]} \tag{1}$$

Unfortunately, error models based only on Pauli noise are not robust enough to describe the real-world noise that must be corrected to realize fault-tolerant quantum computation. [CITATION NEEDED]

Leakage. A leakage error occurs when a qubit "leaks" out of the computational subspace (described by a two-dimensional Hilbert space) into a state that can only be described with a higher-order Hilbert space. If leakage goes undetected, the error compounds over time as it spreads through a multi-qubit interactions. Leakage can be stochastically modeled with a 3-level system (qutrit) with orthonormal basis: $\{\,|0\rangle\,,|1\rangle\,,|2\rangle\,\}$. Leakage events are assumed to be discrete and occur independently with probability p_{leak} at each gate operation acting on the qutrit [3]. The leakage error $E_{E_\uparrow}$ on the jth qutrit in the system can then be understood as:

$$E_{\text{leakage}}(\rho_j) = (1-p_{\text{leak}})\rho_j + p_{\text{leak}}\,|2\rangle\,\langle 2| \qquad \text{[Leakage error channel [3]]} \tag{2}$$

Leakage-reduction units (LRUs) were proposed as a way to convert leakage errors into Pauli-type errors, based on a one-bit teleportation protocol that effectively moves the logical state from the physical qubit in the leaked state to a freshly-prepared physical qubit [3].

All species of qubit experience leakage errors, but it's important to note the underlying *physical* processes look and behave very differently despite sharing a common mathematical description.

Transmon leakage. Transmon single and two qubit gates rely on coherent microwave driving to transition between computational basis states. Two qubit gates such as the controll Z and controll X gates are the primary source of leakage in transmon qubits because of their temporary usage of states outside of the computational basis to induce phase and bit flips in the target qubit. For example, when applying a control Z gate on transmon hardware to the state $|11\rangle$, we temporarily force both excitons onto one qubit, forcing it into the second excited state, $|2\rangle$, and letting the other qubit fall into the ground state, $|0\rangle$, before allowing the system to relax back into $|11\rangle$ with the intended phase shift applied. However, there is a nonzero chance that the system does not relax properly, and leaves one of our qubits in the $|2\rangle$ state, causing a leakage error in our system. **Yang_2024**, **Miao_2023**

Neutral atom / Rydberg leakage. Neutral atom entangling gates exploit Rydberg blockade, which is a quantum phenomenon that occurs when an atom is excited to a particular Rydberg energy level, which prevents nearby atoms from being excited to the same state. The excitation of the single valence electron in alkali metals like Cs, Rb results in a large effective atomic radius [CITATION NEEDED, but I think it's on the order of 25E5], creating a large electric dipole moment.

(one or more electrons in the case of alkaline earth metals or lanthanides like Yb) TODO [4]

Erasure. An erasure error occurs when a qubit is longer detectable by the measurement device, so the quantum state information encoded by the qubit is effectively "lost" or "erased". Mathematically, we extend the qubit Hilbert space with some $|e\rangle$ that is orthogonal to the qubit state $\{|0\rangle, |1\rangle\}$. The erasure channel can be defined as:

$$E_{\text{erase}}(\rho_j) = (1 - p_{\text{erase}})\rho_j + p_{\text{erase}} |e\rangle \langle e| \quad \text{[Erasure error channel [3]]} \tag{3}$$

Photon loss is the dominant form of error in photonic qubits [CITATION NEEDED].

Leakage-detection units (LDUs) were developed to address leakage errors in superconducting qubits, by periodically coupling a data qubit to an acilla that flags when the data qubit leaves the computational subspace. The flagged data qubit can be reset/replaced during subsequent error-correction cycles, effectively treating the flagged qubit as an erasure error.

Neutral atoms / Rydberg arrays, each atom is arranged into a known position using optical tweezers. Although the lattice/trap is situated inside a vacuum chamber, some nonzero amount of background gas collisions or thermal energy transfer from the environment can eject an atom from its lattice site.

Correlated Errors. TODO The most important thing to mention is that real-world errors are often correlated, so fault-tolerant QC requires correction of correlated noise. [5]

Cross-talk might also be worth mentioning for completeness.

2.2 Erasure Qubits

A qubit engineered so the dominant source of errors is from erasures.

A bias towards erasure-type errors occurs natively in some types of qubits (e.g. photons) or can be engineered by converting leakage errors to erasures (e.g. with LDUs in superconducting qubits).

2.2.1 Advantages, Challenges of Erasure Errors

Advantages. From: "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025)

1. Can theoretically correct more erasure errors than unknown Pauli errors. We can correct $d-1$ erasure errors compared to $\frac{d-1}{2}$ Pauli errors. As a result, we expect logical error rate P_L on the order of p^d where p is the physical qubit error rate, compared to $p^{d/2}$ for unknown Pauli errors. [6]

2. Higher thresholds: surface code 50% erasure noise vs. 19% depolarizing noise.

3. Dominant noise source can be engineered

TODO Figure: d=5 surface code with some unknown Pauli error spanning 3 qubits, with syndrome squares highlighted. If we apply the minimum-weight correction, this causes a logical error.

Figure: d=5 surface code with erasure errors, exact locations known.

Challenges. Challenges from "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025):

1. More physical operations.
2. Erasure detection imperfect, introducing another dimension to error models.
3. Midcircuit erasure checks can create correlated noise for neighboring qubits
4. Tools to simulate and decode erasure circuits are active area of research.

Erasure Conversion in Superconducting Qubits

1. Shouzhen Gu. "Fault-tolerant quantum architectures based on erasure qubits" and "Optimizing quantum error correction protocols with erasure qubits" [7] [6]

Erasure Conversion in Neutral / Rydberg Atoms

1. Yue Wu, Shimon Kolkowitz, Shruti Puri, Jeff D Thompson. Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays [8]
2. Gefen Baranes, Madelyn Cain, J. Pablo Bonilla Ataides, Dolev Bluvstein, Josiah Sinclair, Vladan Vuletic, Hengyun Zhou, Mikhail D. Lukin . Leveraging Atom Loss Errors in Fault Tolerant Quantum Algorithms [9]

2.3 Surface Code

Qauntum error correction (QEC) assumes a piece of quantum information is distributed between many entangled physical qubits, realizing a logical qubit with higher fidelity than its constiuent physical qubits. Any errors affecting individual physical qubits can be detected and corrected with an ensemble of stabilizing codes, based on the diagnostic syndrome of the stabilizer code.

TODO surface code explainer TODO Surface code figures

[10]

Circuit-level Noise Thresholds. If physical qubit error is below a critical noise threshold, the logical error rate should be supressed exponentially as the number of physical qubits per logical qubit is increased. In the surface code, the approximate relation between d code distance corresponding to $2d^2 - 1$ physical qubits per logical qubit, the physical error rate p, logical error rate E_d, and error rate threshold p_{thr} [11].

Erasure Tolerance and Enhancements TODO Thresholds for $p_erase << p$, also some useful background on erasure errors in general: (see: Previous Work on Erasure Errors) [12]

$$E_d \propto \frac{p}{p_{thr}}^{(d+1)/2} \qquad \text{[Logical error rate]} \tag{4}$$

Therefore when $p << p_{thr}$, the logical error rate is supressed exponentially with code distance. The exponential supression factor Λ is approximately the linear relation between p_{thr} and p: [11].

$$\Lambda = \frac{E_d}{E_{d+1}} \approx \frac{p_{thr}}{p} \qquad \text{[Supression factor]} \tag{5}$$

2.4 Formalism

define stabilizer ciruits with erasure channel: [6]
compare w/ surface code having only pauli error channel

Simulating Erasure Circuits Describe Stim circuits. Link to Stim circuits / appendix figures of selected timeline-svg operations. [13]

Decoding TODO
Describe the decoding problem for Pauli error stabilizer circuits in the surface code. Explain how surface code error syndromes can be framed as a graph traversal problem with detection events as nodes and errors as the weighted edges. The minimum weight perfect matching (MWPM) algorithm can find the most probable set of errors that occurred. Explain why MWPW works for uniformly random Pauli errors, but why it experiences a sharp drop-off in performace for biased (non-identical) noise models.

Describe the decoding problem when 100 of noise channels can be converted to erasure errors (conversion_ratio=1 in our codebase). Describe how the Hypergraph Minimum-Weight Parity Factor (MWPF). [14]

3 Methods & Results

3.1 Surface Code Thresholds

Circuit Simulation We studied the performance of a rotated surface code under various noise and error syndrome decoding regimes (Table 1), to understand the impact of erasure conversion on the error threshold. We simulate a rotated surface code in Stim [13]. We perform Monte Carlo simulations over a sweep of parameters: code distance $d \in \{3, 5, 7, 9, 11\}$, depolarizing error rate $p \in [0.001, 0.06]$, and erasure error rate $p_e \in [0, 1]$.

We estimate the logical error rate P_L after d rounds of measurements. We assume perfect measurements. Before each round and after each Clifford gate, we model depolarizing noise with uniformly random Pauli errors from the set $\{I, X, Y, Z\}$ In the cases where p_e is nonzero, we convert a percentage of depolarizing errors to erasure errors with a known location. After an erasure error is detected, the affected qubit is reset to a maximally mixed state $I/2$.

The resulting syndrome measurements are decoded to determine whether the sample can be corrected or results in a logical error.

Syndrome decoding The decoding process occurs in two phases:

1. Offline phase (once per unique combination of swept parameters) TODO - explanation of Stim detector error model (DEM), figures generated from DEM. We build a binary parity check matrix H which encodes which error mechanisms flip a particular detector. Each row in H corresponds to a parity check and each column corresponds to some error mechanism. The element $H[i, j]$ is 1 is parity check i is flipped by the error mechanism in j, otherwise $H[i, j]$ is 0.

   The check matrix H can also be understood as a Tanner graph, expressing classical constraints for the error correcting code.

   In the cases where we consider heralded erasure errors, we add a

   Each edge in the graph is weighted by TODO

   TODO For MWPF the row/col representation has multiple incident entries. Cols of not-hyper incidence matrix have exactly two non-zero entries since each edge can only have two vertices (one at each end). The hypergraph representation can have multiple vertices per edge.

2. Online decoding (per shot in Monte Carlo simulation) Sample circuit, measure binary syndrome vector with one bit per detector. PyMatching: sparse blossom to find minimum-weight set of edges w/ incidence matching (exactly) the syndrome. MWPF:

In the cases where p_e is zero (no erasure conversion is performed), we decode the measured error syndrome with a Minimum-Weight Perfect Matching decoder (PyMatching) [15]. TODO - explanation of Stim detector error model (DEM), figures generated from DEM. There is a one-to-one correspondence between each detector in the DEM and a node in the graph passed to PyMatching.

TODO - additionally, we decode using MWPF when $p_e = 0$ as an ablation study.

In the cases where p_e is nonzero, we TODO

ECC	**Decoding Scheme**	p_{thr}	p_e	**Results**
Rotated surface code, X basis	MWPM	0.57 - 0.78	0	Figure 1a, Figure 1b
	MWPF	0.62 - 0.71	0	Figure 1c, Figure 1d
	Hypergraph MWPF	2.70 - 3.02	1	Figure 1e, Figure 1f

Table 1: Summary of error thresholds p_{thr} for various error-correcting codes (ECC), syndrome decoding schemes, and erasure conversion factor p_e.

4 Discussion

TODO why we opted to use mwpf instead of updating PyMatching edge weights, per the Thompson paper / talk.

TODO our opinion of the hypergraph construction of this problem.

5 Conclusion

Acknowledgements Shimon Kolkowitz, Harry Levine, Thiago Bergamaschi

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A Appendix: Syndrome of Loss Errors

Describe how a loss is heralded during state readout. Each potential loss location is described by a distinct error probability, described by a unique configuration of the error syndroms. [EXAMPLE OF ERROR SYNDROME]

B Appendix: Additional Figures and Tables

C Glossary

depolarizing error channel 1

Pauli noise A widely-used error model that characterizes noise as a random combination of Pauli-X, Pauli-Y, and Pauli-Z operations. Many early benchmarks and error correction algorithm thresholds assume an independent and identically distributed Pauli noise model, but real-world error models ar more complex. 1

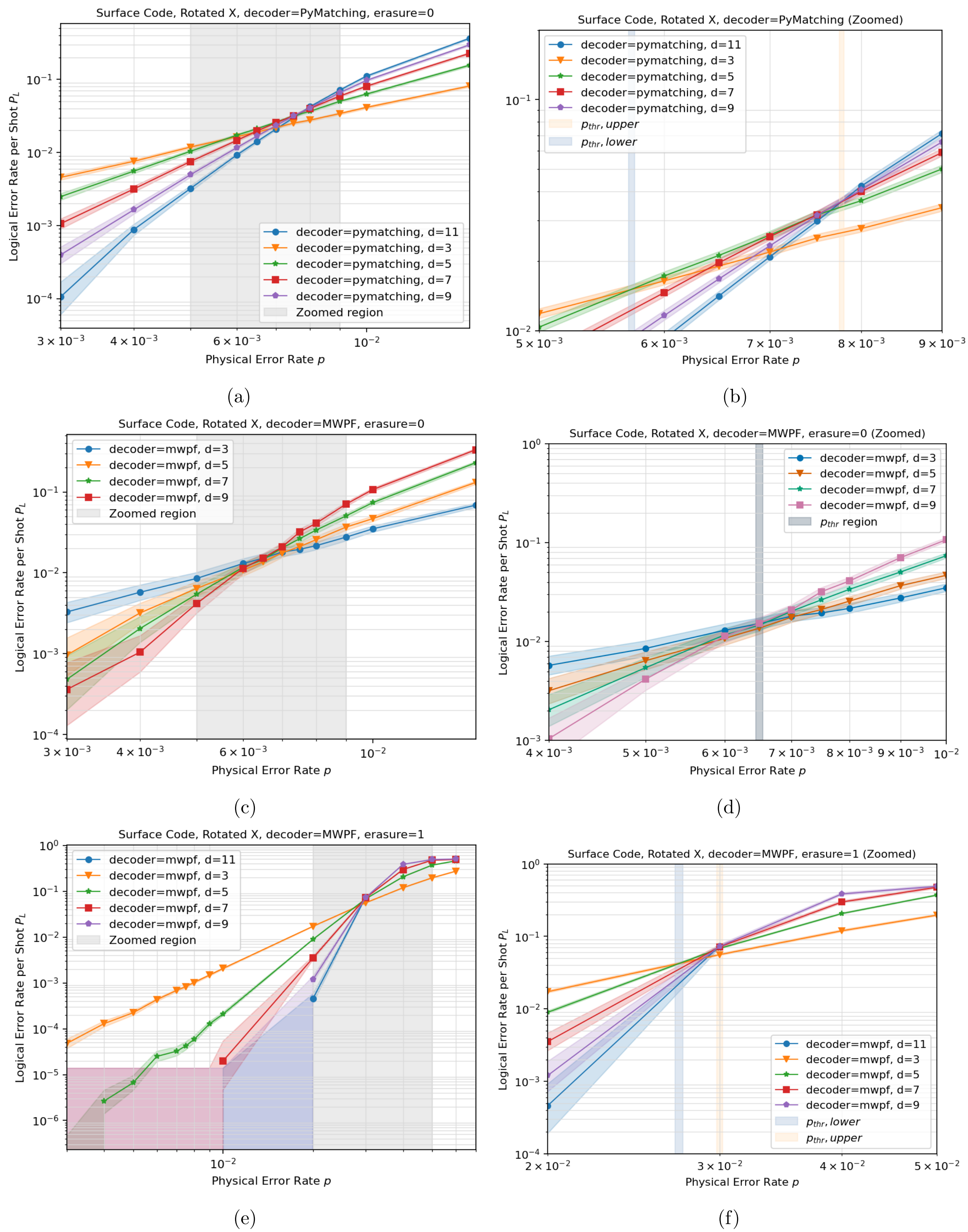

Figure 1: **Circuit-level error thresholds**. Comparison of logical error rate P_L with physical qubit error rate p, for various syndrome decoding schemes and noise regimes. The lower and upper bounds of the threshold region p_{thrs} is estimated from the d=3,5 and d=9,11 crossings respectively. **(a)(b)** Pure depolarizing noise, decoded with Minimum Weight Pefect Matching (PyMatching) with p_{thrs} between 0.57% and 0.78%. **(c)(d)** Pure depolarizing noise, decoded with Minimum Weight Parity Factor, with with p_{thrs} between 0.62% and 0.71%. **(e)(f)**100% of depolarizing noise converted to erasure errors, with p_{thrs} between 2.7% and 3.02%.